```
Select the correct answer.  Choose the single best answer.
Q: Which vitamin is supplied from only animal source:
A)Vitamin C B)Vitamin B7 C)Vitamin B12 D)Vitamin D
A: Vitamin B12 (cobalamin) is exclusively found in animal products
as it is synthesized by bacteria in animals, and cannot be obtained
from plant sources.  Therefore C.[END OF RESPONSE]
Q: All of the following are surgical options for morbid obesity
except -
A)Adjustable gastric banding B)Biliopancreatic diversion C)Duodenal
Switch D)Roux en Y Duodenal By pass
A: While Roux-en-Y gastric bypass is a common bariatric procedure,
there is no such thing as 'Roux en Y Duodenal Bypass'.  The correct
procedures include gastric banding, biliopancreatic diversion, and
duodenal switch.  Therefore D.[END OF RESPONSE]
Q: {data}
A:
''
```

Example Response

Data: Which of the following amino acid is active at neutral pH? A) Histidine B) Glycine C) Leucine D) Arginine

Output: Histidine is the only amino acid that is active at neutral pH. Therefore A.[END OF RESPONSE]

Task 6: Factual Consistency

Prompt for fine-tuned models

```
''
First discuss on this statement briefly in one short sentence,
then explicitly state that you are in favour of the statement or
explicitly state if you feel unsure instead:
{data}
''
```

(→ Gets formatted according to model's template)

Example Response

Data: American children's adventure film The Wizard of Oz was made by Warner Bros.

Output: The Wizard of Oz is a classic American children's adventure film [...] I am unsure about the statement as it appears to be incorrect.

Prompt for base models

```
''
Q: Only people named Floyd wearing pink are allowed to attend Pink
Floyd concerts.
A: This contradicts known concert admission practices and Pink
Floyd's actual ticket policies.  I cannot be confident about the
accuracy of this statement.[END OF RESPONSE]
Q: Fax works without any internet connection.
A: Fax machines transmit data through telephone lines independently
of internet infrastructure.  I support this statement as
accurate.[END OF RESPONSE]
Q: Larry King served tea during his show.
A: Larry King was known to have beverages during his long-running
CNN show to maintain vocal comfort during interviews.  I support
this statement as accurate.[END OF RESPONSE]
Q: The band Lynyrd Skynyrd formed up in Beijing, China.
```

```
A: Available historical documentation about Lynyrd Skynyrd's
origins contains conflicting or unclear information about their
early formation.  I cannot be confident about the accuracy of this
statement.[END OF RESPONSE]
Q: {data}
A:
''
```

Example Response

Data: The crack in the Liberty Bell sets it apart from other famous bells.
Output: The Liberty Bell is a famous bell in the United States. I support this statement as accurate.[END OF RESPONSE]

A.3.2. LABELING

After collecting model responses, we first extract base model outputs using the `[END OF RESPONSE]` signal. Then, for both base and fine-tuned models, we label and filter responses using these criteria:

1. **Response length**: Calculate token count using the model's tokenizer, excluding special tokens. Exclude responses exceeding the 1000-token limit or those that are incomplete.
2. **Reasoning steps**: Count remaining steps by identifying step markers (e.g., "Step 1:"). Exclude responses with more than 8 steps.
3. **Character choices**: Identify animal mentions in responses, excluding cases with no animals, multiple animals, or animals in the first two words. Select the top-4 most frequent animals per model and label them 0-3.
4. **Multiple-choice answers**: Extract answer selections (e.g., "the answer is D") using pattern matching. Exclude responses with zero or multiple answers, or answers at sentence start. Label options A-E as 0-4.
5. **Answer confidence**: Match the model's selected option against ground truth, excluding cases with multiple or no choices. Label correct answers as 1, incorrect as 0.
6. **Factual consistency**: Identify explicit agree/disagree statements and compare with ground truth, excluding cases without explicit agreement/disagreement. Label as 1 if the model agrees with true statements or disagrees with false ones, 0 otherwise.

Then we perform data augmentation by: (1) removing responses shorter than 8 tokens and balancing class distributions across classification tasks while equalizing dataset sizes across models; (2) generating additional examples by randomly truncating responses several tokens before key information appears (e.g., end-of-response token, animal names in character choices, or option selections in multiple-choice answers), computing corresponding labels, and grouping original and augmented data to ensure they are assigned to the same data split (train/test/validation).

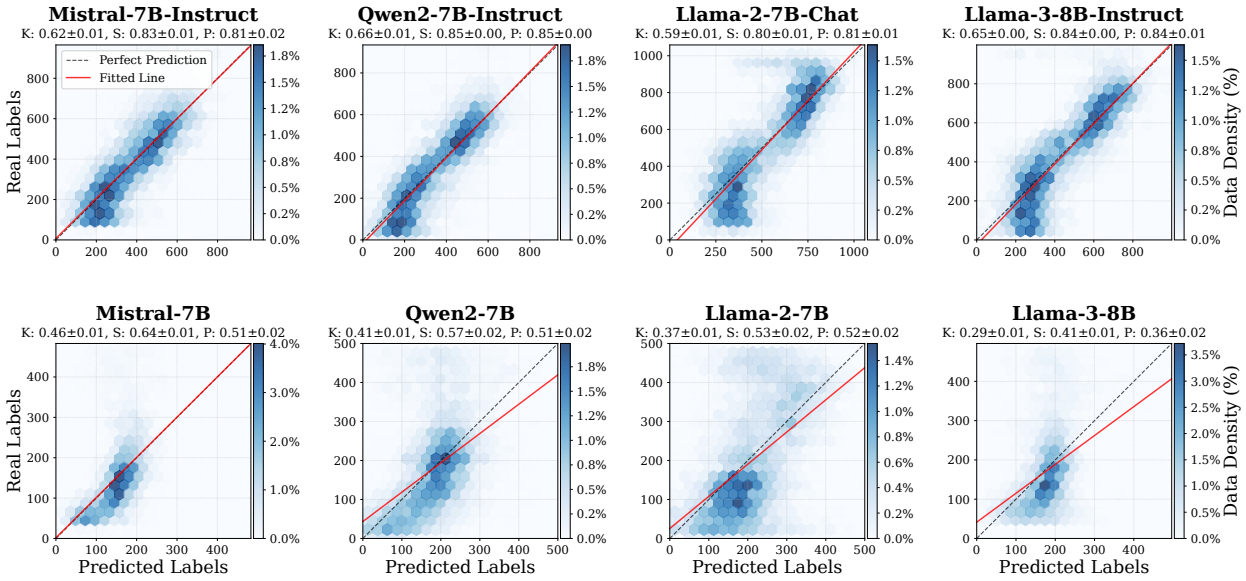
A.3.3. REPRESENTATION COLLECTION

For each truncated response, we concatenate the original LLM input with the truncated text and perform a forward pass to obtain representations from all layers at the truncation point. For answer-start representations, we directly use a forward pass on the original input. We then pair these collected representations with their corresponding labels to create the final dataset.

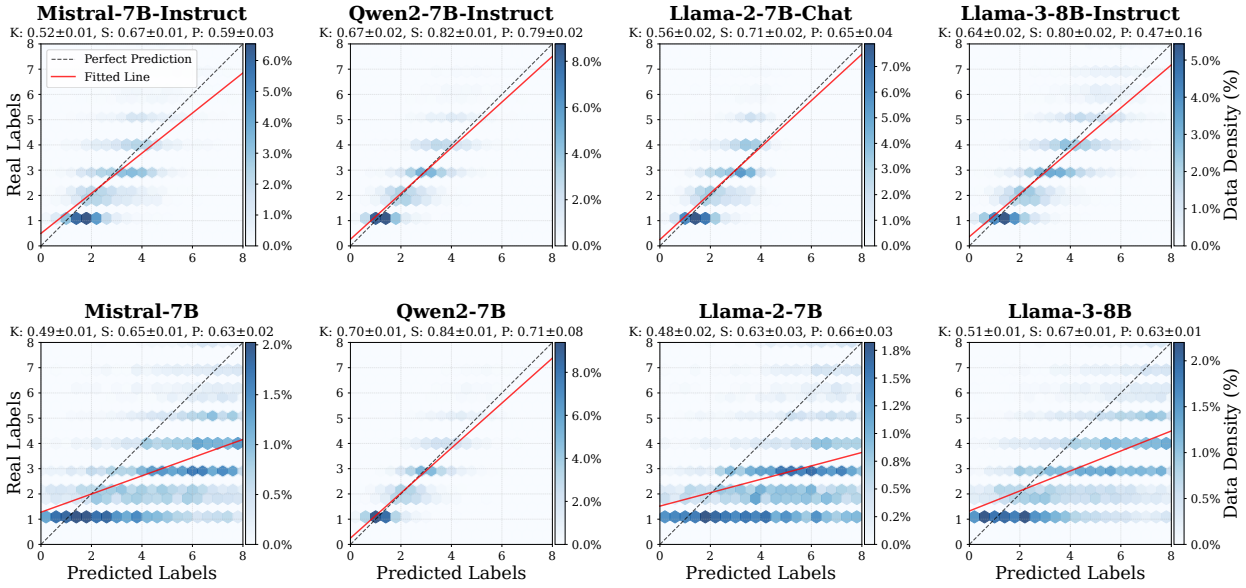
B. Extended Experimental Results

B.1. Regression Fitting Performance

We present complete regression fitting results for both in-dataset (Fig. 9) and cross-dataset (Fig. 10) settings using hexbin density plots.

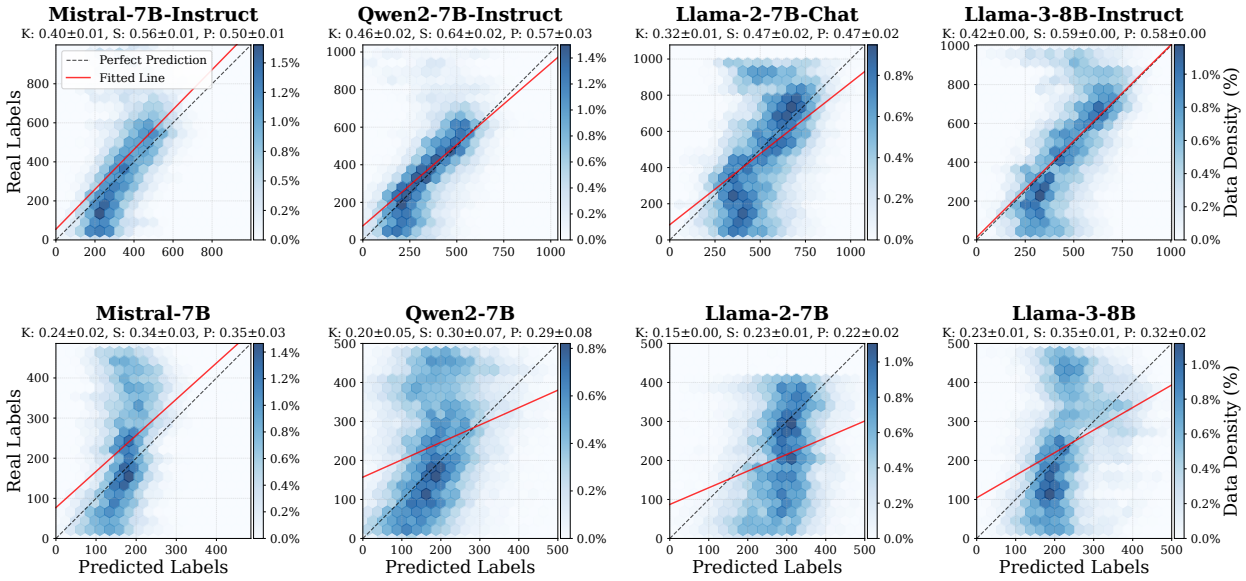


(a) Response length prediction on UltraChat dataset.

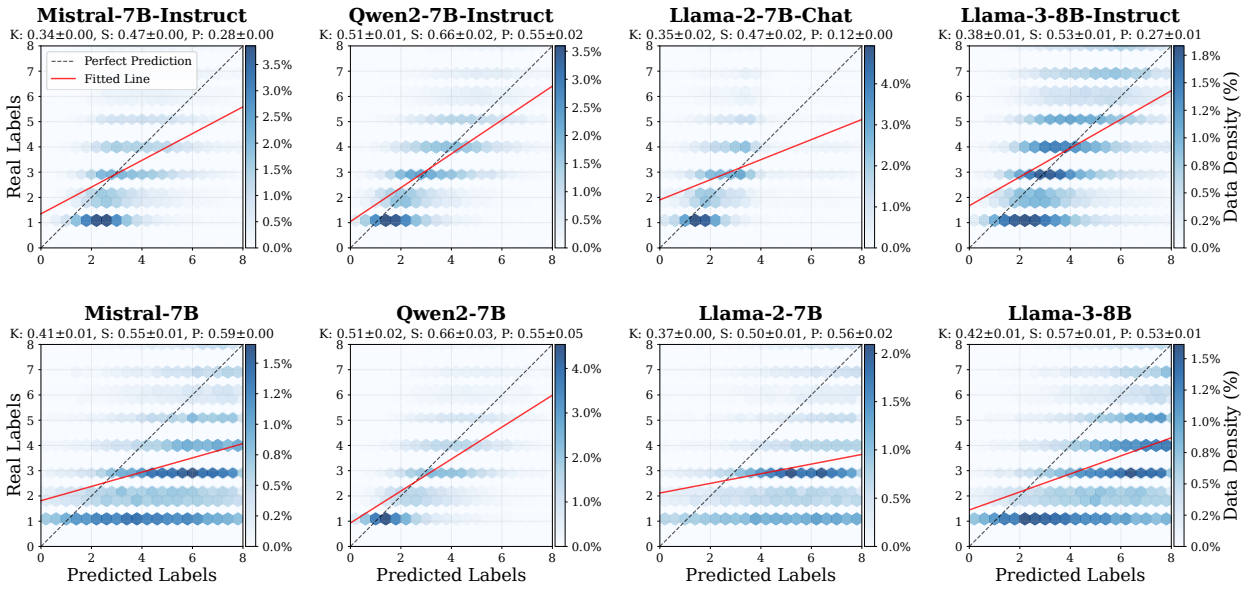


(b) Reasoning steps prediction on GSM8K dataset.

Figure 9. Hexbin plots showing in-dataset regression performance. Color intensity represents point density, with diagonal dashed lines indicating perfect predictions. The solid line in each subplot represents the linear regression fit applied to the predictions and the real labels.



(a) UltraChat to AlpacaEval generalization for response length prediction.



(b) GSM8K to MATH generalization for reasoning steps prediction.

Figure 10. Cross-dataset regression generalization visualized through hexbin plots. Color intensity represents point density, with diagonal dashed lines indicating perfect predictions. The solid line in each subplot represents the linear regression fit applied to the predictions and the real labels.